



Elevated loss sensitivity in the reward circuit in adolescents with video game but not social media addiction

Xu He^{a,b}, Yu Chen^b, Wei Zhang^{a,*}, Chiang-Shan R. Li^{b,c,d,**}

^a School of Psychology, South China Normal University, Guangzhou, 510631, China

^b Department of Psychiatry, Yale University School of Medicine, New Haven, CT, 06520, USA

^c Department of Neuroscience, Yale University School of Medicine, New Haven, CT, 06520, USA

^d Wu Tsai Institute, Yale University, New Haven, CT, 06520, USA

ARTICLE INFO

Handling Editor: Catalina L. Toma

Keywords:

Video game addiction
Social media addiction
Longitudinal data
Reward circuit
Loss sensitivity

ABSTRACT

Both video game addiction (VGA) and social media addiction (SMA) have been linked to dysfunction of the brain reward circuit. However, it remains unclear whether VGA or SMA have a bidirectional relationship with reward circuit dysfunction during development. The current study used a large longitudinal dataset from the Adolescent Brain and Cognitive Development (ABCD) study to explore the association between VGA, SMA, and longitudinal changes in neural processing of rewards and losses during a monetary incentive delay task. Significant VGA $\times$ time interactions were observed for loss-related neural activity were observed in left nucleus accumbens, bilateral insula, and right anterior cingulate cortex (ACC). Simple slope analysis revealed an increasing trend in neural responses to losses among participants with high levels of VGA. Additionally, time $\times$ neural loss sensitivity interactions predicted later VGA in the right amygdala and right ACC, suggesting that heightened loss sensitivity both influences and is influenced by VGA. In contrast, SMA showed no significant longitudinal associations with reward or loss sensitivity. These findings highlight a bidirectional relationship between VGA and neural loss sensitivity, while SMA was not implicated in similar patterns. These findings may provide valuable insights into the neural mechanisms underlying behavioral addiction.

1. Introduction

Smart devices such as smartphones and tablets has enabled and raised concerns about the dependent use of the Internet among adolescents (Lozano-Blasco et al., 2022; Mohammad et al., 2023; Sun & Zhang, 2021). In fact, Internet-related addiction has significantly increased over the past two decades (Meng et al., 2022), with prevalence rates reaching 17.42% for social media addiction (SMA) and 6.04% for video game addiction (VGA). Converging evidence demonstrates that addictive use of video games or social media is associated with adverse psychosocial consequences among adolescents (Ahmed et al., 2022; Arora et al., 2021; Boer et al., 2021). A meta-analysis revealed that adolescents engaging in VGA had a 1.57 times higher probability of developing mental health problems, including depression, anxiety, and emotional distress, compared to those without such behavior (Purwaningsih & Nurmala, 2021). Similar links have also been observed for SMA (Huang, 2022). Therefore, it is imperative to investigate the mechanisms to

inform the pathophysiology and to advance prevention and early intervention of the adverse consequences of SMA and VGA.

The Interaction of Person-Affect-Cognition-Execution (I-PACE) model of Internet addiction (Brand et al., 2016, 2019) conceptualizes both VGA and SMA as specific types of Internet addiction, characterized by symptoms such as obsessive thoughts, tolerance, diminished impulse control, and withdrawal. Within this framework, diminished executive functioning and heightened reward-seeking tendencies interact to foster a craving to go online, thereby encouraging maladaptive decision-making. Despite the potential for long-term harm, individuals seeking immediate gratification or emotional relief may persist in these behaviors, ultimately leading to compulsive engagement. While the I-PACE model outlines general mechanisms driving Internet addiction, it also underscores the importance of distinguishing specific types of Internet addiction, including VGA and SMA. Empirical evidence supports this differentiation (Pontes, 2017). For example, Montag et al. (2015) identified only moderate correlation between VGA and SMA,

* Corresponding author. School of Psychology, South China Normal University, Guangzhou, Guangdong, 510631, China.

** Corresponding author. Department of Psychiatry, Yale University School of Medicine, New Haven, CT, 06519, USA.

E-mail addresses: zhangwei@scnu.edu.cn (W. Zhang), chiang-shan.li@yale.edu (C.-S.R. Li).

<https://doi.org/10.1016/j.chb.2025.108554>

Received 16 September 2024; Received in revised form 14 December 2024; Accepted 1 January 2025

Available online 2 January 2025

0747-5632/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

emphasizing the need to study these behaviors separately. [Burén et al. \(2021\)](#) reported a stronger association between VGA and males, while SMA symptoms showed stronger correlations with psychosomatic problems. Additionally, a systematic review by [Wegmann and Brand \(2020\)](#) highlighted more consistent evidence linking VGA to impairments in executive functions and decision-making compared to SMA.

Despite growing interest in potentially differentiable pathophysiology of VGA and SMA ([Morris et al., 2023](#); [Wacks & Weinstein, 2021](#)), studies directly their underlying neural mechanisms remain scarce. [Dieter et al. \(2017\)](#) conducted the only study comparing individuals with VGA, SMA, and healthy controls on social anxiety-related inhibitory control using an Emotional Stroop Task. The results revealed significant hypoactivations in the left middle and superior temporal gyrus in the VGA group when processing interference from socially anxious words, as compared to the SMA group. This finding highlights the potential neural differentiation between VGA and SMA but also underscores the need for further research. Given the central role of reward processing—including reward-seeking and gratification—in the development and maintenance of VGA and SMA ([Brand et al., 2016](#)), it is equally crucial to investigate the neural substrates of these processes. The current study aims to examine how VGA and SMA may be distinguished in the neural correlates of reward processing and to contribute to the emerging literature and a more comprehensive framework for refining theoretical models of Internet addiction, such as I-PACE.

The reward circuit, comprising limbic, prefrontal, and striatal regions in the brain ([Arias-Carrión et al., 2010](#); [Cooper et al., 2017](#); [Elliott et al., 2000](#)), governs reward-seeking and punishment avoidance. Dysfunction within this circuit has been associated with VGA ([Fujiwara et al., 2022](#); [Wang, 2023](#)). For example, [Yao et al. \(2020\)](#) reported that individuals with VGA as compared to healthy controls exhibited blunted striatal activity when experiencing losses. Other studies have reported differing results, with VGA linked to diminished neural response to gains but not losses ([Li et al., 2020](#)) or reduced activation to both gains and losses ([Dong et al., 2017](#)). Resting-state functional magnetic resonance imaging (fMRI) studies have further implicated frontal-striatal circuit dysfunction in VGA ([Von Deneen et al., 2022](#); [Yan et al., 2021](#)). In contrast, research on reward circuit dysfunction in SMA remains limited. [Meshi et al. \(2013\)](#) demonstrated that nucleus accumbens activity in response to social rewards could predict Facebook usage intensity. Additionally, reduced gray matter volumes in the amygdala have been associated with SMA ([He et al., 2017](#)).

Building on the evidence linking VGA and SMA to reward circuit dysfunction, the current study employs the Monetary Incentive Delay Task (MID; [Knutson et al., 2001](#)), a widely used paradigm for investigating reward-related neural activity. The MID task consistently identifies key brain regions involved in reward processing, including the striatum, insula, and amygdala during reward and/or loss anticipation, as well as the anterior cingulate cortex (ACC), orbitofrontal cortex (OFC), and ventromedial prefrontal cortex (vmPFC) during reward and/or loss receipt ([Chen et al., 2022](#); [Oldham et al., 2018](#)). Additionally, studies of the MID task suggest that individuals with addictive behaviors exhibit significant differences in striatal activity during the anticipatory phase of reward processing, whereas such differences are less pronounced during the outcome phase ([Balodis & Potenza, 2015](#)). These findings align with broader evidence suggesting that altered reward processing may play a critical role in the neural mechanisms underlying VGA and SMA. The current study, therefore, focuses on examining frontal-striatal regions implicated in the MID task, including the nucleus accumbens, amygdala, caudate, putamen, insula, ACC, OFC, and vmPFC, to further elucidate the neural mechanisms underlying VGA and SMA.

To move beyond cross-sectional insights and explore the temporal dynamics of neural processing in VGA and SMA, this study uses a large longitudinal dataset from the Adolescent Brain and Cognitive Development study (ABCD; [Casey et al., 2018](#)) to investigate changes in reward-related brain activity. Specifically, the study examines how VGA

and SMA are associated with neural processing of rewards and losses during the MID task over time. Two key questions guide this research: First, the study tests whether VGA and SMA have long-term effects on neural sensitivity to rewards and losses. It is hypothesized that adolescents with higher levels of VGA or SMA will exhibit reduced neural responses during reward and loss processing. Second, the study explores whether changes in neural processing of rewards and losses precede the onset of VGA and SMA. Here, it is hypothesized that lower neural sensitivity to rewards and losses predicts higher levels of VGA and SMA over time.

2. Methods

2.1. Participants

The ABCD Study is an ongoing longitudinal study, encompassing nearly 12,000 participants aged 9–10 years at baseline, drawn from 21 sites across the United States. It provides a valuable dataset to understand brain function and behavior during adolescent development. The ABCD study received approval from the Centralized Institutional Review Board at the University of California, San Diego (IRB# 160091) and the institutional review boards of each individual site.

Participants in the ABCD study underwent clinical and behavioral assessment every year and brain imaging every other year. The VGA and SMA questionnaires were introduced in the ABCD dataset starting from the second year ([Bagot et al., 2022](#)). Thus, the current study included two waves of data collected each at year 2 and 4 with both the needed behavioral and imaging measures. Participants were excluded if they met any of the following criteria at any time point: 1) lacking basic demographic data (i.e., sex, age, race/ethnicity, and family income); 2) lacking VGA or SMA data; 3) lacking fMRI data for the MID task; or 4) imaging data suggested for exclusion by the ABCD analytic core ([Hagler et al., 2019](#)). Based on the specified criteria, 2400 participants met the inclusion criteria in the second year, and 1578 met the criteria in the fourth year. However, not all participants from the year 4 also participated in the year 2, resulting in a final sample of 543 participants (including 212 females) who met the criteria at both time points. No significant demographic differences were found between the final sample and the original group of 10,964 participants. Demographic

Table 1
Demographic information.

	Original sample N = 10964		Final sample N = 543	
	N	%	N	%
Age				
11	5709	52.1	244	44.9
12	5229	47.7	298	54.9
13	26	0.2	1	0.2
Sex				
Male	5753	52.4	331	61.0
Female	5217	47.5	212	39.0
Race/ethnicity				
White	5882	53.6	261	48.1
Black	1571	14.3	80	14.7
Hispanic	2164	19.7	132	24.3
Asian	207	1.9	10	1.8
Other	1147	10.5	60	11.0
Family income				
Less than \$5000	346	3.2	19	3.5
\$5000 to \$11,999	364	3.3	25	4.6
\$12,000 to \$15,999	228	2.1	19	3.5
\$16,000 to \$24,999	462	4.2	29	5.3
\$25,000 to \$34,999	598	5.4	42	7.7
\$35,000 to \$49,999	861	7.8	68	12.5
\$50,000 to \$74,999	1388	12.6	93	17.1
\$75,000 to \$99,999	1505	13.7	84	15.5
\$100,000 to \$199,999	3168	28.9	121	22.3
\$200,000 and higher	1175	10.7	43	7.9

information of this cohort was presented in Table 1.

2.2. Measures

SMA and VGA. The Social Media Addiction Questionnaire is a 6-item questionnaire designed to assess addictive behaviors related to social media use, with example items like: “I’ve become stressed or upset if I am not allowed to use my social media apps.” The Video Game Addiction Questionnaire is a 6-item questionnaire to assess addictive use of video game, with example statements include: “I play video games so I can forget about my problems.”. These questionnaires were adapted from the Bergen Facebook Addiction Scale (Andreassen et al., 2012), and all items were rated on a Likert scale ranging from 1 (Never) to 6 (Very often). These questionnaires were administered only to a subset of participants who reported having at least one social media account or engaging in video game use. The reliability indexed by McDonald’s ω was 0.86 for SMA and 0.90 for VGA.

MID Task. The MID task was used in the ABCD study (Casey et al., 2018) to evaluate neural activation to rewards and losses. Each trial of the MID task presented participants with an incentive cue representing one of five possible outcomes (small win: \$0.20; large win: \$5; small loss: \$0.20; large loss: \$5; neutral outcome: \$0 at stake; Fig. 1). After an anticipation interval ranging from 1500 to 4000 ms, participants have to respond to a target (150–500 ms) as fast as possible to win rewards or avoid losses. Then, participants received a feedback informing them of the outcome following the button press. Each participant completed two runs of the MID task, with each run comprising 50 trials arranged in a pseudo-random order. Task difficulty is dynamically adjusted during the task to achieve a 60% accuracy rate. If the participant’s accuracy across the previous six trials fail to meet a specified threshold, the target duration is extended. Conversely, if the accuracy exceeds the threshold, the target duration is reduced. The MID task allows for the identification

Task-related activation was modeled using a gamma variate basis function with its temporal derivative, incorporating stimulus timing for each condition to compute linear contrasts (e.g., reward vs. negative feedback) at the individual level. The resulting outputs included beta coefficients and standard errors for each ROI, which were averaged across runs for each participant. The current analysis utilized average beta weights for two specific contrasts—Reward Anticipation vs. Neutral (reward contrast) and Loss Anticipation vs. Neutral (loss contrast)—across eight predefined ROIs, which were provided in the curated dataset (see data dictionary, <https://data-dict.abcdstudy.org/>). These ROIs correspond to the following parcellations from the data: nucleus accumbens (accumbens area), amygdala (amygdala), caudate (caudate), putamen (putamen), insula (insula), ACC (caudal anterior cingulate), OFC (medial orbitofrontal), and vmPFC (rostral anterior cingulate). Further details regarding fMRI preprocessing and quality control procedures can be found in Hagler et al. (2019).

Covariates. Age, sex, race/ethnicity, and family income were included as covariates in the analyses. Age measured in years was used as a continuous variable. Sex was treated as a dichotomous variable, coded as 1 for male and 2 for female. Race/ethnicity was collapsed into five levels: White, Black, Hispanic, Asian, and Other. Family income was rated on a 10-point scale ranging from 1 (Less than \$5000) to 10 (\$200,000 and greater).

2.3. Statistical analyses

We employed the linear mixed models (LMMs) in the R lme4 package (Bates et al., 2015) to examine the interaction effects of VGA and SMA by time on neural responses to rewards and losses.

To investigate the long-term effects of VGA and SMA on neural reward/loss sensitivity (hypothesis 1), we specified Model 1 as follows:

$$\text{Beta Weights (for the reward / loss contrasts in each ROI)} \sim \text{VGA} + \text{SMA} + \text{Time} + \text{VGA} \times \text{Time} + \text{SMA} \times \text{Time} + \text{Age} + \text{Sex} \\ + \text{Race/Ethnicity} + \text{Family income} + \text{Mean framewise displacement} + (1 \mid \text{Site})$$

of neural activation associated with rewards and losses.

fMRI Data Processing. The fMRI data were analyzed at the individual subject level using a general linear model and a region of interest (ROI)-based approach. Following preprocessing, average time courses were extracted for anatomically-defined cortical and subcortical ROIs.

Time refers to data collection conducted at two time points: the two-year follow-up (Time 1, ages 11–13 years) and the four-year follow-up (Time 2, ages 13–15 years). Beta-weights for the reward/loss contrasts in each ROI served as dependent variables in separate models. VGA score, SMA score, Time, and their interaction as independent variables.

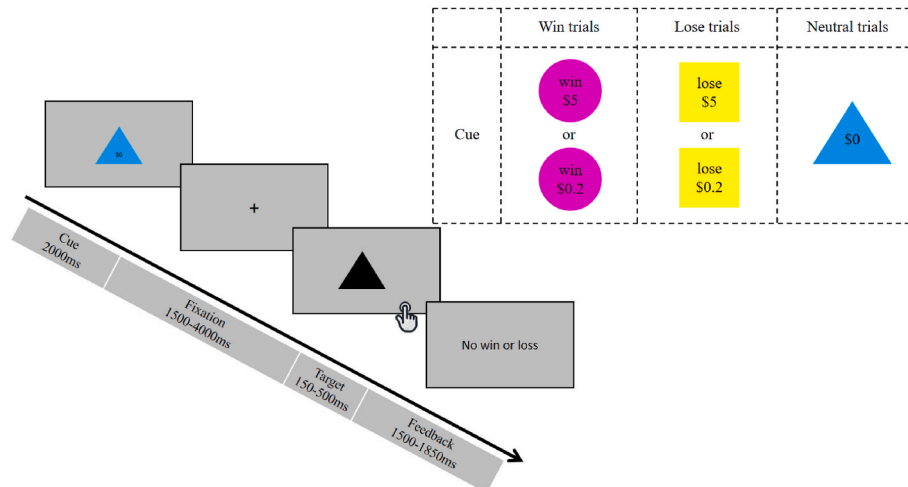


Fig. 1. Monetary incentive delay task.

The interactions of $VGA \times \text{Time}$ and $SMA \times \text{Time}$ were included to determine if the effects of VGA and SMA on neural responses changed over time, beyond the main effects. Age, sex, race/ethnicity, and family income were included as fixed-effect covariates to control for potential confounding influences, and research sites were modeled as random effects to account for the nested data structure. We also included mean framewise displacement during fMRI scanning as a covariate to control for head motion. All models were corrected using the Benjamini-Hochberg method (Benjamini & Hochberg, 1995).

To test whether changes in neural processing of rewards and losses predict the increase in severity of VGA and SMA (hypothesis 2), we employed a second set of LMMs:

$$\begin{aligned} \text{VGA or SMA} \sim & \text{Beta Weights} + \text{Time} + \text{Beta Weights} * \text{Time} + \text{Age} \\ & + \text{Sex} + \text{Race/Ethnicity} + \text{Family income} \\ & + \text{Mean Framewise Displacement} + (1 | \text{Site}) \end{aligned}$$

Here, VGA score and SMA score were the dependent variables, while beta-weights for the reward/loss contrasts in each ROI, Time, and their interaction were the independent variables in separate models. Age, sex, race/ethnicity, and family income were again included as fixed-effect covariates, and research sites were modeled as random effects.

After identifying a significant interaction (e.g., $VGA \times \text{time}$), we categorized participants into three subgroups based on their VGA scores (-1 SD, mean, and $+1$ SD). For each subgroup, we then fitted separate models that included site as a random effect and retained the same covariates as in the original model. This approach maintains consistency in controlling for confounding variables and provides a methodologically rigorous means of examining how the interaction differs across varying levels of VGA.

3. Results

3.1. General results

Fig. 2 shows the distribution of VGA and SMA scores at Time 1 and Time 2. Paired-samples t -test revealed that VGA and SMA scores were significantly lower at Time 1 compared to Time 2 (VGA: $t = -2.80$, $p = .005$, Cohen's $d = -0.12$; SMA: $t = -5.34$, $p < .001$, Cohen's $d = -0.23$). Beta-weights for the reward contrast in bilateral nucleus accumbens were significantly higher at Time 1 compared to Time 2 (Left nucleus accumbens: $t = 2.33$, $p = .020$, Cohen's $d = 0.10$; SMA: $t = 2.64$, $p = .008$, Cohen's $d = 0.11$). The differences in beta-weights for the reward

contrast in other ROIs or for the loss contrast between Time 1 and Time 2 did not reach statistical significance ($ps > 0.05$).

There was a moderate correlation between VGA and SMA scores (Time 1: $r = 0.426$, $p < .001$; Time 2: $r = 0.500$, $p < .001$; see Table 2) after controlling for age, sex, race/ethnicity, and family income. At Time 1, the VGA score was negatively correlated with beta-weights for the loss contrast in right insula ($r = -0.085$, $p = .049$) and left ACC ($r = -0.089$, $p = .039$), while the SMA score was positively correlated with beta-weights for the loss contrast in right insula ($r = 0.100$, $p = .020$). Conversely, at Time 2, the VGA score exhibited positive correlations with beta-weights for the reward contrast in bilateral amygdala (left: $r = 0.092$, $p = .038$; right: $r = 0.122$, $p = .004$), as well as for the loss contrast in right amygdala ($r = 0.137$, $p = .001$) and right ACC ($r = 0.088$, $p = .042$). The SMA score showed positive correlations with beta-weights for the reward contrast in left amygdala ($r = 0.089$, $p = .038$) and for the loss contrast in bilateral amygdala (left: $r = 0.119$, $p = .006$; right: $r = 0.085$, $p = .047$).

3.2. The interaction effects of $VGA \times \text{time}$ and $SMA \times \text{time}$ on neural loss sensitivity

Significant interaction effects of VGA and time on neural responses to losses were observed in left nucleus accumbens ($b = 0.106$, $SE = 0.034$, 95% CI [0.040, 0.172], $p = .002$, $p_{FDR} = 0.032$), left insula ($b = 0.103$, $SE = 0.033$, 95% CI [0.037, 0.168], $p = .002$, $p_{FDR} = 0.032$), right insula ($b = 0.111$, $SE = 0.034$, 95% CI [0.045, 0.177], $p = .001$, $p_{FDR} = 0.032$), and right ACC ($b = 0.111$, $SE = 0.034$, 95% CI [0.048, 0.180], $p = .001$, $p_{FDR} = 0.032$; see Table 3 for the outcomes of the LMMs and see Fig. 3 for the brain regions showing significant effects). Simple slope analysis revealed a decreasing trend in loss sensitivity among participants with lower VGA scores (-1 SD), while those with higher VGA ($+1$ SD) scores exhibited increased neural responses to losses in these brain regions. No significant interaction effects of SMA and time on neural reward or loss sensitivity were observed in any of the ROIs (p_{FDR} 's > 0.05). Additionally, the main effect of SMA also did not reach the significant level (p_{FDR} 's > 0.05).

3.3. Test of the interaction effects of neural reward/loss sensitivity $\times$ time on VGA and SMA

Significant interaction effects of neural loss sensitivity and time on VGA were observed in right amygdala ($b = 0.086$, $SE = 0.029$, 95% CI [0.029, 0.144], $p = .003$, $p_{FDR} = 0.048$) and right ACC ($b = 0.088$, $SE = 0.030$, 95% CI [0.029, 0.147], $p = .003$, $p_{FDR} = 0.048$; see Table 4). Simple slope analysis revealed that participants with lower neural loss sensitivity (-1 SD) showed decreased levels of VGA, while those with higher neural loss sensitivity ($+1$ SD) exhibited increased levels of VGA. No significant interaction on SMA were observed (p_{FDR} 's > 0.05).

4. Discussion

The current study tested the association between VGA, SMA, and the neural reward/loss sensitivity at two time point. Through a longitudinal analysis, a significant $VGA \times \text{time}$ interaction was observed, indicating that variations in VGA over time are associated with corresponding changes in neural loss sensitivity. Specifically, participants with lower VGA scores exhibited a decreasing sensitivity to losses, whereas those with higher VGA scores showed elevated neural responses to losses in the left nucleus accumbens, left insula, right insula, and right ACC. Moreover, participants displaying higher neural loss sensitivity in the right amygdala and right ACC subsequently reported increased levels of VGA. In contrast, SMA did not demonstrate any long-term effects on neural sensitivity to rewards or losses in the examined ROIs. Taken together, these findings suggested that a bidirectional relationship between VGA and elevated neural sensitivity to losses.

Despite a negative correlation at time 1, adolescents with higher

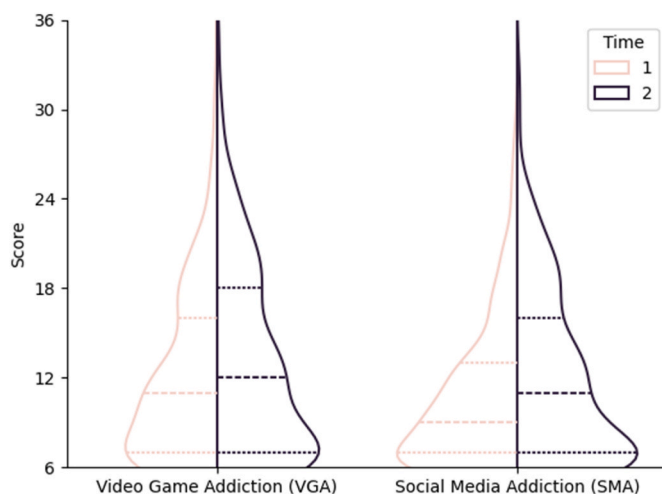


Fig. 2. The distribution of Video Game Addiction (VGA) and Social Media Addiction (SMA) scores at Time 1 (two-year follow-up) and Time 2 (four-year follow-up).

Table 2
Correlations between Video Game Addiction (VGA), Social Media Addiction (SMA), and brain measures at Time 1 (two-year follow-up) and Time 2 (four-year follow-up).

		Time 1				Time 2			
		VGA		SMA		VGA		SMA	
		<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>
Time 1	VGA	–							
	SMA	0.426	<0.001	–					
Time 2	VGA	0.410	<0.001	0.215	<0.001	–			
	SMA	0.275	<0.001	0.394	<0.001	0.500	<0.001	–	
Beta-weights for the reward contrast									
Nucleus accumbens	L	–0.007	0.869	0.013	0.768	0.050	0.251	–0.023	0.589
	R	–0.012	0.787	0.040	0.359	0.014	0.746	–0.015	0.728
Amygdala	L	0.003	0.937	–0.014	0.754	0.092	0.032	0.089	0.038
	R	–0.018	0.674	–0.011	0.795	0.122	0.004	0.076	0.079
Caudate	L	–0.058	0.180	0.027	0.529	0.041	0.345	–0.022	0.610
	R	–0.055	0.205	0.015	0.735	0.044	0.306	–0.019	0.653
Putamen	L	–0.047	0.272	0.008	0.852	0.010	0.814	–0.035	0.413
	R	–0.052	0.226	–0.002	0.968	0.010	0.817	–0.021	0.628
Insula	L	–0.040	0.350	0.042	0.329	0.041	0.344	0.020	0.651
	R	–0.073	0.088	0.008	0.844	0.044	0.306	0.023	0.592
ACC	L	–0.084	0.051	–0.012	0.776	0.023	0.598	–0.030	0.485
	R	–0.058	0.176	0.019	0.654	0.037	0.389	–0.019	0.652
OFC	L	–0.048	0.263	–0.043	0.324	0.057	0.186	0.028	0.510
	R	–0.016	0.707	–0.033	0.446	0.025	0.560	–0.010	0.812
vmPFC	L	–0.001	0.985	0.033	0.444	0.034	0.432	–0.013	0.755
	R	–0.005	0.910	0.010	0.821	0.039	0.370	0.015	0.728
Beta-weights for the loss contrast									
NAc	L	–0.083	0.053	0.026	0.541	0.065	0.132	–0.003	0.943
	R	–0.054	0.210	0.059	0.168	0.075	0.081	0.001	0.982
Amygdala	L	0.012	0.775	–0.010	0.814	0.077	0.074	0.119	0.006
	R	–0.032	0.461	0.026	0.551	0.137	0.001	0.085	0.047
Caudate	L	–0.063	0.142	0.059	0.173	0.042	0.327	0.014	0.752
	R	–0.057	0.184	0.061	0.158	0.066	0.124	0.024	0.583
Putamen	L	–0.058	0.176	0.055	0.204	0.056	0.196	0.036	0.402
	R	–0.070	0.104	0.060	0.167	0.040	0.356	0.019	0.653
Insula	L	–0.074	0.086	0.066	0.129	0.078	0.072	0.059	0.172
	R	–0.085	.049	0.100	.020	0.078	0.072	0.034	0.438
ACC	L	–0.089	.039	0.019	0.664	0.042	0.334	0.010	0.808
	R	–0.082	0.058	0.051	0.237	0.088	0.042	0.040	0.356
OFC	L	–0.061	0.155	–0.008	0.853	0.023	0.597	0.029	0.499
	R	–0.060	0.166	0.006	0.898	0.030	0.481	–0.002	0.963
vmPFC	L	–0.080	0.062	0.045	0.296	0.042	0.327	0.033	0.443
	R	–0.070	0.103	0.037	0.392	0.043	0.317	0.055	0.205

Note. Conditioned on age, sex, race/ethnicity, and family income. ACC, anterior cingulate cortex; OFC, orbitofrontal cortex; vmPFC, ventromedial prefrontal cortex; L, left hemisphere; R, right hemisphere.

VGA scores showed increased loss sensitivity across time. Conversely, those with heightened loss sensitivity exhibited elevated levels of VGA, indicating a mutually reinforcing cycle. The findings is contrary to previous cross-sectional studies involving young adults (Raiha et al., 2020), which indicated blunted sensitivity to losses in relation to VGA. One possible reason for these conflicting results is the ongoing developmental changes in brain regions involved in reward and loss processing during adolescence. For example, a developmental fMRI study found a quadratic effect of age on loss processing in the anterior INS, with a significant decrease during mid-adolescence (Insel & Somerville, 2018). Similarly, other studies have documented changes in reward sensitivity across development, showing a peak of reward activation in nucleus accumbens (Schreuders et al., 2018). These findings suggest that VGA may affect the normal trajectory of neurodevelopmental changes in reward and loss processing during adolescence, which need to be further tested with a multi-wave longitudinal data.

The results demonstrated a significant impact of VGA on loss sensitivity in the left nucleus accumbens, left insula, right insula, and right ACC, highlighting the importance of these brain regions in development of VGA. This pattern suggests that the neural mechanisms involved in processing negative feedback and guiding behavioral adjustments may be disrupted. For instance, heightened activity in the nucleus accumbens is associated not only with reward-related processes but also with aversive behaviorss (Klawonn & Malenka, 2018), while increased insula

activation may correspond to more intense emotional responses to losses (Uddin et al., 2017), such as stress or frustration, that inadvertently encourage continued gaming. Similarly, changes in ACC activation may signal difficulties in recruiting cognitive control or adapting strategies (Gasquoine, 2013), resulting in persistent gaming behaviors. Establishing these links between neural function and observable behavior can inform preventive strategies and interventions focused on enhancing emotion regulation and decision-making skills in adolescents with VGA. Approaches such as cognitive-behavioral therapy, neuromodulation techniques, or digital interventions that promote healthier gaming habits might be informed by a deeper understanding of the specific neural circuits involved.

The results showed a positive correlation between SMA and loss sensitivity at both time 1 and time 2, as well as reward sensitivity at time 2. However, the long-term impact of SMA on reward/loss sensitivity did not reach statistical significance. Notably, a recent finding has identified distinct developmental trajectory in brain responsiveness to social feedback between adolescents with low and high SMA (Flannery et al., 2024). Specifically, the low SMA group exhibited increasing sensitivity to positive social feedback across puberty, whereas the high SMA group demonstrated elevated social feedback sensitivity only at the onset of puberty. These findings suggest that adolescents with high SMA may prioritize addiction-related rewards (e.g., social feedback) over monetary rewards. Therefore, future studies should take the characteristics of

Table 3

Interaction effects of Video Game Addiction (VGA) and Social Media Addiction (SMA) by time on beta-weights for reward and loss contrast in each ROI.

ROI		VGA × Time				SMA × Time			
		<i>b</i>	95% CI	<i>p</i>	<i>p</i> _{FDR}	<i>b</i>	95% CI	<i>p</i>	<i>p</i> _{FDR}
Beta-weights for the reward contrast									
Nucleus accumbens	L	0.05	[-0.02, 0.12]	0.147	0.254	−0.039	[-0.11, 0.03]	0.248	0.333
	R	0.02	[-0.05, 0.08]	0.588	0.660	−0.028	[-0.10, 0.04]	0.404	0.491
Amygdala	L	0.05	[-0.02, 0.11]	0.160	0.259	0.014	[-0.05, 0.08]	0.674	0.719
	R	0.07	[0.01, 0.14]	0.031	0.104	0.006	[-0.06, 0.07]	0.857	0.871
Caudate	L	0.08	[0.01, 0.14]	0.023	0.090	−0.058	[-0.12, 0.01]	0.091	0.182
	R	0.08	[0.01, 0.14]	0.021	0.090	−0.051	[-0.12, 0.02]	0.131	0.238
Putamen	L	0.06	[0.00, 0.13]	0.062	0.180	−0.051	[-0.12, 0.02]	0.134	0.238
	R	0.06	[-0.01, 0.13]	0.077	0.182	−0.037	[-0.10, 0.03]	0.281	0.367
Insula	L	0.07	[0.00, 0.13]	0.047	0.150	−0.047	[-0.11, 0.02]	0.164	0.259
	R	0.08	[0.01, 0.14]	0.021	0.090	−0.028	[-0.10, 0.04]	0.407	0.491
ACC	L	0.08	[0.02, 0.15]	0.013	0.083	−0.042	[-0.11, 0.02]	0.213	0.302
	R	0.09	[0.02, 0.15]	0.011	0.078	−0.061	[-0.13, 0.01]	0.073	0.182
OFC	L	0.05	[-0.02, 0.11]	0.160	0.259	0.015	[-0.05, 0.08]	0.662	0.718
	R	0.03	[-0.03, 0.1]	0.341	0.436	−0.012	[-0.08, 0.06]	0.729	0.753
vmPFC	L	0.04	[-0.03, 0.11]	0.250	0.333	−0.047	[-0.11, 0.02]	0.166	0.259
	R	0.03	[-0.04, 0.09]	0.433	0.513	−0.022	[-0.09, 0.05]	0.519	0.604
Beta-weights for the loss contrast									
Nucleus accumbens	L	0.11	[0.04, 0.17]	0.002	0.032	−0.061	[-0.13, 0.01]	0.073	0.182
	R	0.08	[0.01, 0.14]	0.024	0.090	−0.059	[-0.13, 0.01]	0.085	0.182
Amygdala	L	0.02	[-0.05, 0.08]	0.657	0.718	0.042	[-0.03, 0.11]	0.217	0.302
	R	0.09	[0.03, 0.16]	0.006	0.055	−0.013	[-0.08, 0.05]	0.695	0.729
Caudate	L	0.07	[0.01, 0.14]	0.028	0.100	−0.053	[-0.12, 0.01]	0.120	0.226
	R	0.08	[0.01, 0.15]	0.018	0.089	−0.054	[-0.12, 0.01]	0.114	0.221
Putamen	L	0.09	[0.02, 0.16]	0.009	0.072	−0.058	[-0.13, 0.01]	0.087	0.182
	R	0.09	[0.03, 0.16]	0.006	0.055	−0.066	[-0.13, 0.00]	0.052	0.158
Insula	L	0.10	[0.04, 0.17]	0.002	0.032	−0.057	[-0.12, 0.01]	0.090	0.182
	R	0.11	[0.05, 0.18]	0.001	0.032	−0.081	[-0.15, −0.02]	0.017	0.089
ACC	L	0.09	[0.03, 0.16]	0.006	0.055	−0.045	[-0.11, 0.02]	0.187	0.278
	R	0.11	[0.05, 0.18]	0.001	0.032	−0.061	[-0.13, 0.01]	0.070	0.182
OFC	L	0.04	[-0.02, 0.11]	0.213	0.302	0.002	[-0.07, 0.07]	0.949	0.949
	R	0.06	[-0.01, 0.12]	0.088	0.182	−0.03	[-0.10, 0.04]	0.369	0.463
vmPFC	L	0.08	[0.01, 0.15]	0.018	0.089	−0.046	[-0.11, 0.02]	0.181	0.276
	R	0.06	[-0.01, 0.13]	0.081	0.182	−0.021	[-0.09, 0.05]	0.532	0.608

Note. The table reports standardized beta coefficients, 95% CI, uncorrected and corrected *p* values. * refers to effects that were significant survived Benjamini-Hochberg correction ($p_{FDR} < 0.05$). Age, sex, race/ethnicity, family income, and mean framewise displacement were included as covariates. The research sites were modeled as random effects to account for the nested structure of the data. ACC, anterior cingulate cortex; OFC, orbitofrontal cortex; vmPFC, ventromedial prefrontal cortex; L, left hemisphere; R, right hemisphere.

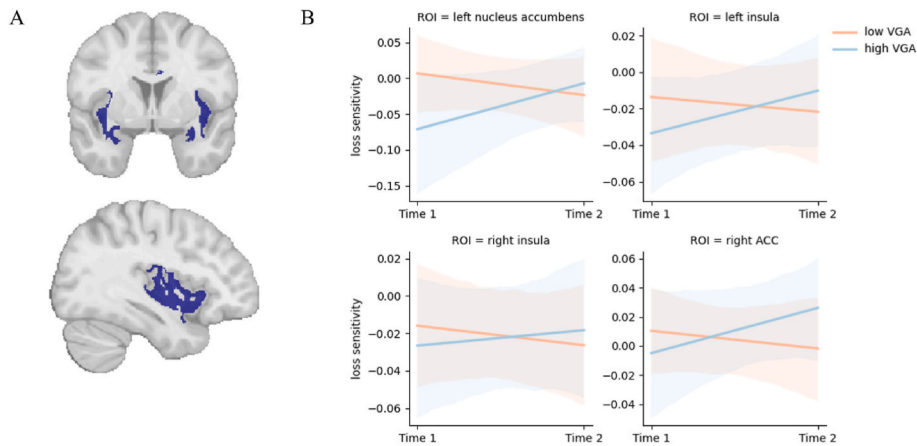


Fig. 3. Significant interaction effects of Video Game Addiction (VGA) and Time on loss sensitivity. **A.** Brain regions showing significant effects, including left nucleus accumbens, right ACC (anterior cingulate cortex), and bilateral insula. **B.** Temporal trends in loss sensitivity across four brain regions, comparing participants with low (-1 SD) versus high ($+1$ SD) VGA.

specific types of Internet addiction into consideration to explore the neurodevelopmental outcomes.

Several limitations of the current study should also be acknowledged. Firstly, the inclusion of only two time points limits the ability to establish a causal relationship between VGA and loss sensitivity. Additional time points of the ABCD data will be included in our future study

to replicate the findings when available, thereby providing a more robust understanding of the long-term effects. Secondly, the overall levels of VGA and SMA in the sample are relatively low. While this may lead to an underestimation of the effects of VGA and SMA on neurodevelopment, it may also enhance the generalizability of the findings to a broader population. Thirdly, several mental health disorders,

Table 4

Interaction effects of neural reward/loss sensitivity and time on Video Game Addiction (VGA).

Interactions	ROI		VGA			
			<i>b</i>	95% CI	<i>p</i>	<i>p</i> _{FDR}
Beta-weights for reward contrast × Time	Nucleus accumbens	L	0.04	[-0.02, 0.10]	0.231	0.296
		R	0.02	[-0.04, 0.08]	0.606	0.606
	Amygdala	L	0.06	[0.00, 0.11]	0.065	0.136
		R	0.07	[0.01, 0.13]	0.020	0.107
	Caudate	L	0.06	[0.00, 0.12]	0.057	0.136
		R	0.06	[0.00, 0.11]	0.068	0.136
	Putamen	L	0.03	[-0.03, 0.09]	0.324	0.363
		R	0.03	[-0.03, 0.09]	0.329	0.363
	Insula	L	0.05	[-0.01, 0.11]	0.091	0.152
		R	0.06	[0.00, 0.12]	0.039	0.131
	ACC	L	0.05	[-0.01, 0.11]	0.084	0.152
		R	0.05	[-0.01, 0.11]	0.100	0.152
	OFC	L	0.05	[-0.01, 0.10]	0.123	0.179
		R	0.02	[-0.04, 0.08]	0.423	0.437
	vmPFC	L	0.03	[-0.03, 0.09]	0.357	0.381
		R	0.03	[-0.03, 0.09]	0.257	0.305
Beta-weights for loss contrast × Time	Nucleus accumbens	L	0.07	[0.01, 0.13]	0.018	0.107
		R	0.07	[0.01, 0.13]	0.029	0.131
	Amygdala	L	0.05	[-0.02, 0.11]	0.150	0.200
		R	0.09	[0.03, 0.14]	0.003	0.048
	Caudate	L	0.05	[-0.01, 0.11]	0.087	0.152
		R	0.06	[0.00, 0.12]	0.039	0.131
	Putamen	L	0.06	[0.00, 0.12]	0.068	0.136
		R	0.05	[-0.01, 0.11]	0.095	0.152
	Insula	L	0.08	[0.02, 0.14]	0.008	0.064
		R	0.08	[0.02, 0.14]	0.008	0.064
	ACC	L	0.06	[0.00, 0.12]	0.041	0.131
		R	0.09	[0.03, 0.15]	0.003	0.048
	OFC	L	0.03	[-0.02, 0.09]	0.252	0.305
		R	0.04	[-0.01, 0.10]	0.136	0.189
	vmPFC	L	0.06	[0.00, 0.12]	0.048	0.136
		R	0.06	[0.00, 0.12]	0.059	0.136

Note. The table reports standardized beta coefficients, 95% CI, uncorrected and corrected *p* values. * refers to effects that were significant survived Benjamini-Hochberg correction ($p_{FDR} < 0.05$). Age, sex, race/ethnicity, family income, and mean framewise displacement were included as covariates. The research sites were modeled as random effects to account for the nested structure of the data. ACC, anterior cingulate cortex; OFC, orbitofrontal cortex; vmPFC, ventromedial prefrontal cortex; L, left hemisphere; R, right hemisphere.

including depression and anxiety, are known to be associated with VGA and SMA but were not considered in this study. Future research should explore these psychological factors to better understand their interactions with addictive behaviors. Lastly, the effect sizes observed in the current results are modest. However, the large sample size and the longitudinal design have the potential to reveal significant impacts when considering the cumulative effects during adolescence. Therefore, further research should focus on the possible amplification or attenuation of these effects over time.

5. Conclusion

Using longitudinal data from the ABCD Study, the current study examined the long-term effects of VGA and SMA on neural responses to rewards and losses. The findings highlight distinct mechanisms: VGA exhibited a bidirectional relationship with heightened loss sensitivity, suggesting that altered loss processing both contributes to and results from addictive gaming behaviors. In contrast, SMA showed no comparable long-term association with reward or loss sensitivity. These results underscore the importance of considering the unique developmental trajectories of VGA and SMA, providing valuable insights for future research and interventions targeting adolescent behavioral addictions.

CRedit authorship contribution statement

Xu He: Writing – original draft, Formal analysis, Conceptualization. **Yu Chen:** Writing – review & editing, Validation. **Wei Zhang:** Writing – review & editing, Conceptualization. **Chiang-Shan R. Li:** Writing – review & editing, Conceptualization.

Funding

This work was supported by the National Natural Science Foundation of China [grant numbers 32271115].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Data used in the preparation of this article were obtained from the Adolescent Brain Cognitive DevelopmentSM (ABCD) Study (<https://abcdstudy.org>), held in the NIMH Data Archive (NDA). This is a multisite, longitudinal study designed to recruit more than 10,000 children age 9–10 and follow them over 10 years into early adulthood. The ABCD Study® is supported by the National Institutes of Health and additional federal partners under award numbers U01DA041048, U01DA050989, U01DA051016, U01DA041022, U01DA051018, U01DA051037, U01DA050987, U01DA041174, U01DA041106, U01DA041117, U01DA041028, U01DA041134, U01DA050988, U01DA051039, U01DA041156, U01DA041025, U01DA041120, U01DA051038, U01DA041148, U01DA041093, U01DA041089, U24DA041123, U24DA041147. A full list of supporters is available at <https://abcdstudy.org/federal-partners.html>. A listing of participating sites and a complete listing of the study investigators can be found at https://abcdstudy.org/consortium_members/. ABCD consortium investigators designed and implemented the study and/or provided data but did not necessarily participate in the analysis or writing of this report. This manuscript reflects the views of the authors and may not reflect the opinions or views of the NIH or ABCD consortium investigators. The ABCD data repository grows and changes over time. The ABCD data used in this report came from <https://nda.nih.gov/study.html?id=2313>. DOIs can be found at 10.15154/z563-zd24.

Data availability

The authors do not have permission to share data.

References

- Ahmed, G. K., Abdalla, A. A., Mohamed, A. M., Mohamed, L. A., & Shamaa, H. A. (2022). Relation between internet gaming addiction and comorbid psychiatric disorders and emotion avoidance among adolescents: A cross-sectional study. *Psychiatry Research*, 312, Article 114584. <https://doi.org/10.1016/j.psychres.2022.114584>
- Andreasen, C. S., Torsheim, T., Brunborg, G. S., & Pallesen, S. (2012). Development of a Facebook addiction scale. *Psychological Reports*, 110(2), 501–517. <https://doi.org/10.2466/02.09.18.PR0.110.2.501-517>
- Arias-Carrión, O., Stamelou, M., Murillo-Rodríguez, E., Menéndez-González, M., & Pöppel, E. (2010). Dopaminergic reward system: A short integrative review. *International Archives of Medicine*, 3(1), 24. <https://doi.org/10.1186/1755-7682-3-24>
- Arora, A., Chakraborty, P., & Bhatia, M. P. S. (2021). Problematic use of digital technologies and its impact on mental health during COVID-19 pandemic: Assessment using machine learning. In I. Arpacı, M. Al-Emran, M. A. Al-Sharafi, & G. Marques (Eds.), *Emerging technologies during the era of COVID-19 pandemic* (Vol. 348, pp. 197–221). Springer International Publishing. https://doi.org/10.1007/978-3-030-67716-9_13
- Bagot, K., Tomko, R., Marshall, A. T., Hermann, J., Cummins, K., Ksinan, A., Kakalis, M., Breslin, F., Lisdahl, K., Mason, M., Redhead, J., Squeglia, L., Thompson, W., Wade, T., Tapert, S., Fuemmeler, B., & Baker, F. (2022). Youth screen use in the ABCD® study. *Developmental Cognitive Neuroscience*, 57, Article 101150. <https://doi.org/10.1016/j.dcn.2022.101150>
- Balodis, I. M., & Potenza, M. N. (2015). Anticipatory reward processing in addicted populations: A focus on the monetary incentive delay task. *Biological Psychiatry*, 77(5), 434–444. <https://doi.org/10.1016/j.biopsych.2014.08.020>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using *lme4*. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society - Series B: Statistical Methodology*, 57(1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Boer, M., Stevens, G. W. J. M., Finkenauer, C., De Looze, M. E., & Van Den Eijnden, R. J. J. M. (2021). Social media use intensity, social media use problems, and mental health among adolescents: Investigating directionality and mediating processes. *Computers in Human Behavior*, 116, Article 106645. <https://doi.org/10.1016/j.chb.2020.106645>
- Brand, M., Wegmann, E., Stark, R., Müller, A., Wölfling, K., Robbins, T. W., & Potenza, M. N. (2019). The Interaction of Person-Affect-Cognition-Execution (I-PACE) model for addictive behaviors: Update, generalization to addictive behaviors beyond internet-use disorders, and specification of the process character of addictive behaviors. *Neuroscience & Biobehavioral Reviews*, 104, 1–10. <https://doi.org/10.1016/j.neubiorev.2019.06.032>
- Brand, M., Young, K. S., Laier, C., Wölfling, K., & Potenza, M. N. (2016). Integrating psychological and neurobiological considerations regarding the development and maintenance of specific Internet-use disorders: An Interaction of Person-Affect-Cognition-Execution (I-PACE) model. *Neuroscience & Biobehavioral Reviews*, 71, 252–266. <https://doi.org/10.1016/j.neubiorev.2016.08.033>
- Burén, J., Nutley, S. B., Sandberg, D., Ström Wiman, J., & Thorell, L. B. (2021). Gaming and social media addiction in university students: Sex differences, suitability of symptoms, and association with psychosocial difficulties. *Frontiers in Psychiatry*, 12, Article 740867. <https://doi.org/10.3389/fpsy.2021.740867>
- Casey, B. J., Cannonier, T., Conley, M. I., Cohen, A. O., Barch, D. M., Heitzeg, M. M., Soules, M. E., Teslovich, T., Dellarco, D. V., Garavan, H., Orr, C. A., Wager, T. D., Banich, M. T., Speer, N. K., Sutherland, M. T., Riedel, M. C., Dick, A. S., Bjork, J. M., Thomas, K. M., ... Dale, A. M. (2018). The adolescent brain cognitive development (ABCD) study: Imaging acquisition across 21 sites. *Developmental Cognitive Neuroscience*, 32, 43–54. <https://doi.org/10.1016/j.dcn.2018.03.001>
- Chen, Y., Chaudhary, S., & Li, C.-S. R. (2022). Shared and distinct neural activity during anticipation and outcome of win and loss: A meta-analysis of the monetary incentive delay task. *NeuroImage*, 264, Article 119764. <https://doi.org/10.1016/j.neuroimage.2022.119764>
- Cooper, S., Robison, A. J., & Mazei-Robison, M. S. (2017). Reward circuitry in addiction. *Neurotherapeutics*, 14(3), 687–697. <https://doi.org/10.1007/s13311-017-0525-z>
- Dieter, J., Hoffmann, S., Mier, D., Reinhard, I., Beutel, M., Vollstädt-Klein, S., Kiefer, F., Mann, K., & Leménager, T. (2017). The role of emotional inhibitory control in specific internet addiction – an fMRI study. *Behavioural Brain Research*, 324, 1–14. <https://doi.org/10.1016/j.bbr.2017.01.046>
- Dong, G., Li, H., Wang, L., & Potenza, M. N. (2017). Cognitive control and reward/loss processing in Internet gaming disorder: Results from a comparison with recreational Internet game-users. *European Psychiatry*, 44, 30–38. <https://doi.org/10.1016/j.eurpsy.2017.03.004>
- Elliott, R., Friston, K. J., & Dolan, R. J. (2000). Dissociable neural responses in human reward systems. *Journal of Neuroscience*, 20(16), 6159–6165. <https://doi.org/10.1523/JNEUROSCI.20-16-06159.2000>
- Flannery, J. S., Burnell, K., Kwon, S.-J., Jorgensen, N. A., Prinstein, M. J., Lindquist, K. A., & Telzer, E. H. (2024). Developmental changes in brain function linked with addiction-like social media use two years later. *Social Cognitive and Affective Neuroscience*, 19(1), nsae008. <https://doi.org/10.1093/scan/nsae008>
- Fujiwara, H., Tsurumi, K., Shibata, M., Kobayashi, K., Miyagi, T., Ueno, T., Oishi, N., & Murai, T. (2022). Life habits and mental health: Behavioural addiction, health benefits of daily habits, and the reward system. *Frontiers in Psychiatry*, 13, Article 813507. <https://doi.org/10.3389/fpsy.2022.813507>
- Gasquoine, P. G. (2013). Localization of function in anterior cingulate cortex: From psychosurgery to functional neuroimaging. *Neuroscience & Biobehavioral Reviews*, 37(3), 340–348. <https://doi.org/10.1016/j.neubiorev.2013.01.002>
- Hagler, D. J., Hatton, S. N., Cornejo, M. D., Makowski, C., Fair, D. A., Dick, A. S., Sutherland, M. T., Casey, B. J., Barch, D. M., Harms, M. P., Watts, R., Bjork, J. M., Garavan, H. P., Hilmer, L., Pung, C. J., Scat, C. S., Kuperman, J., Bartsch, H., Xue, F., ... Dale, A. M. (2019). Image processing and analysis methods for the adolescent brain cognitive development study. *NeuroImage*, 202, Article 116091. <https://doi.org/10.1016/j.neuroimage.2019.116091>
- He, Q., Turel, O., & Bechara, A. (2017). Brain anatomy alterations associated with Social Networking Site (SNS) addiction. *Scientific Reports*, 7(1), Article 45064. <https://doi.org/10.1038/srep45064>
- Huang, C. (2022). A meta-analysis of the problematic social media use and mental health. *International Journal of Social Psychiatry*, 68(1), 12–33. <https://doi.org/10.1177/0020764020978434>
- Insel, C., & Somerville, L. H. (2018). Asymmetric neural tracking of gain and loss magnitude during adolescence. *Social Cognitive and Affective Neuroscience*, 13(8), 785–796. <https://doi.org/10.1093/scan/nsy058>
- Klawonn, A. M., & Malenka, R. C. (2018). Nucleus accumbens modulation in reward and aversion. *Cold Spring Harbor Symposia on Quantitative Biology*, 83, 119–129. <https://doi.org/10.1101/sqb.2018.83.037457>
- Knutson, B., Adams, C. M., Fong, G. W., & Hommer, D. (2001). Anticipation of increasing monetary reward selectively recruits nucleus accumbens. *Journal of Neuroscience*, 21(16), Article RC159. <https://doi.org/10.1523/JNEUROSCI.21-16-j0002.2001>
- Li, Q., Wang, Y., Yang, Z., Dai, W., Zheng, Y., Sun, Y., & Liu, X. (2020). Dysfunctional cognitive control and reward processing in adolescents with Internet gaming disorder. *Psychophysiology*, 57(2), Article e13469. <https://doi.org/10.1111/psyp.13469>
- Lozano-Blasco, R., Robres, A. Q., & Sánchez, A. S. (2022). Internet addiction in young adults: A meta-analysis and systematic review. *Computers in Human Behavior*, 130, Article 107201. <https://doi.org/10.1016/j.chb.2022.107201>
- Meng, S.-Q., Cheng, J.-L., Li, Y.-Y., Yang, X.-Q., Zheng, J.-W., Chang, X.-W., Shi, Y., Chen, Y., Lu, L., Sun, Y., Bao, Y.-P., & Shi, J. (2022). Global prevalence of digital addiction in general population: A systematic review and meta-analysis. *Clinical Psychology Review*, 92, Article 102128. <https://doi.org/10.1016/j.cpr.2022.102128>
- Meshi, D., Morawetz, C., & Heekeren, H. R. (2013). Nucleus accumbens response to gains in reputation for the self relative to gains for others predicts social media use. *Frontiers in Human Neuroscience*, 7, 439. <https://doi.org/10.3389/fnhum.2013.00439>
- Mohammad, S., Jan, R. A., & Alsaedi, S. L. (2023). Symptoms, mechanisms, and treatments of video game addiction. *Cureus*, 15(3), Article e36957. <https://doi.org/10.7759/cureus.36957>
- Montag, C., Bey, K., Sha, P., Li, M., Chen, Y., Liu, W., Zhu, Y., Li, C., Markett, S., Keiper, J., & Reuter, M. (2015). Is it meaningful to distinguish between generalized and specific internet addiction? Evidence from a cross-cultural study from Germany, Sweden, Taiwan and China. *Asia-Pacific Psychiatry*, 7(1), 20–26. <https://doi.org/10.1111/appy.12122>
- Morris, R., Moretta, T., & Potenza, M. N. (2023). The psychobiology of problematic use of social media. *Current Behavioral Neuroscience Reports*, 10(4), 65–74. <https://doi.org/10.1007/s40473-023-00261-8>
- Oldham, S., Murawski, C., Fornito, A., Youssef, G., Yücel, M., & Lorenzetti, V. (2018). The anticipation and outcome phases of reward and loss processing: A neuroimaging meta-analysis of the monetary incentive delay task. *Human Brain Mapping*, 39(8), 3398–3418. <https://doi.org/10.1002/hbm.24184>
- Pontes, H. M. (2017). Investigating the differential effects of social networking site addiction and Internet gaming disorder on psychological health. *Journal of Behavioral Addictions*, 6(4), 601–610. <https://doi.org/10.1556/2006.6.2017.075>
- Purwaningsih, E., & Nurmala, I. (2021). The impact of online game addiction on adolescent mental health: A systematic review and meta-analysis. *Open Access Macedonian Journal of Medical Sciences*, 9(F), 260–274. <https://doi.org/10.3889/oamjms.2021.6234>
- Raiha, S., Yang, G., Wang, L., Dai, W., Wu, H., Meng, G., Zhong, B., & Liu, X. (2020). Altered reward processing system in internet gaming disorder. *Frontiers in Psychiatry*, 11, Article 599141. <https://doi.org/10.3389/fpsy.2020.599141>
- Schreuders, E., Braams, B. R., Blankenstein, N. E., Peper, J. S., Güroğlu, B., & Crone, E. A. (2018). Contributions of reward sensitivity to ventral striatum activity across adolescence and early adulthood. *Child Development*, 89(3), 797–810. <https://doi.org/10.1111/cdev.13056>
- Sun, Y., & Zhang, Y. (2021). A review of theories and models applied in studies of social media addiction and implications for future research. *Addictive Behaviors*, 114, Article 106699. <https://doi.org/10.1016/j.addbeh.2020.106699>
- Uddin, L. Q., Nomi, J. S., Hébert-Seropian, B., Ghaziri, J., & Boucher, O. (2017). Structure and function of the human insula. *Journal of Clinical Neurophysiology*, 34(4), 300–306. <https://doi.org/10.1097/WNP.0000000000000377>
- Von Deneen, K. M., Hussain, H., Waheed, J., Xinwen, W., Yu, D., & Yuan, K. (2022). Comparison of frontostriatal circuits in adolescent nicotine addiction and internet gaming disorder. *Journal of Behavioral Addictions*, 11(1), 26–39. <https://doi.org/10.1556/2006.2021.00086>
- Wacks, Y., & Weinstein, A. M. (2021). Excessive smartphone use is associated with health problems in adolescents and young adults. *Frontiers in Psychiatry*, 12, Article 669042. <https://doi.org/10.3389/fpsy.2021.669042>

- Wang, F. (2023). Comparison of neural circuitry mechanisms between internet game addiction and heroin addiction. *Science Insights*, 42(3), 857–866. <https://doi.org/10.15354/si.23.re249>
- Wegmann, E., & Brand, M. (2020). Cognitive correlates in gaming disorder and social networks use disorder: A comparison. *Current Addiction Reports*, 7(3), 356–364. <https://doi.org/10.1007/s40429-020-00314-y>
- Yan, H., Li, Q., Yu, K., & Zhao, G. (2021). Large-scale network dysfunction in youths with internet gaming disorder: A meta-analysis of resting-state functional connectivity studies. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 109, Article 110242. <https://doi.org/10.1016/j.pnpbp.2021.110242>
- Yao, Y.-W., Liu, L., Worhunsky, P. D., Lichenstein, S., Ma, S.-S., Zhu, L., Shi, X.-H., Yang, S., Zhang, J.-T., & Yip, S. W. (2020). Is monetary reward processing altered in drug-naïve youth with a behavioral addiction? Findings from internet gaming disorder. *NeuroImage: Clinical*, 26, Article 102202. <https://doi.org/10.1016/j.nicl.2020.102202>